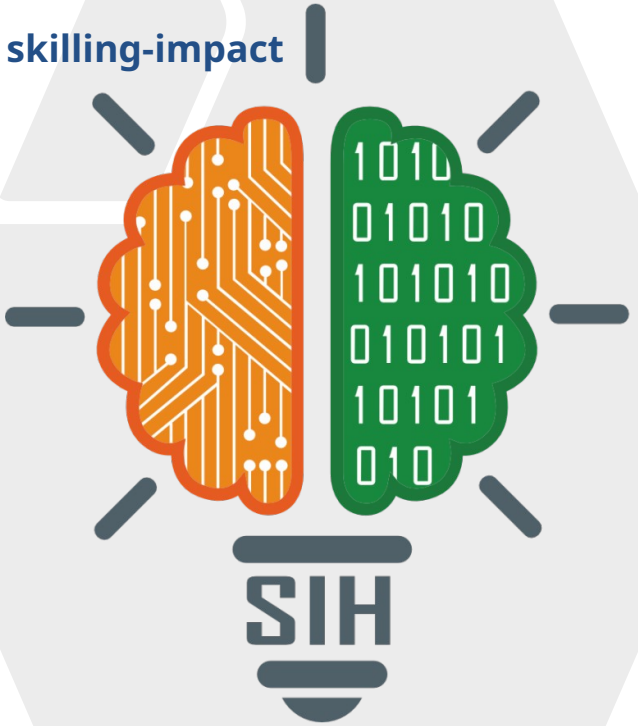
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SKILL PULSE

AI-powered skill, employment & skilling-impact intelligence



- **Problem Statement ID:** SIH26135
- **Problem Statement Title:** Difficulties in tracking employment outcomes, skill gaps, and the impact of skilling initiatives
- **Theme:** Miscellaneous
- **PS Category:** Software
- **Team ID:** _____
- **Team Name:** BuildX

BuildX

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SKILL PULSE

THE PROBLEM → THE FIX

One intelligence layer for the skilling lifecycle

Training
enrolment ≠ outcome

Skills
gaps stay hidden

Jobs
demand is fragmented

Impact
hard to measure

SKILL PULSE ENGINE

Outcome Pulse 01
Track placement, retention & wage signals*

Skill Gap Radar 02
Current skills ↔️ market-required skills

Training Impact 03
Connect training to skill & outcome evidence

What-If Simulator 04
Compare baseline vs intervention scenarios

Unique: person-level intelligence + regional intelligence + policy simulation

*Outcome tracking is the deployment layer; current prototype demonstrates the intelligence workflow.

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TECHNICAL APPROACH

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SYSTEM WORKFLOW

PROFILE
location • qualification • profession • skills

NORMALIZE
skill aliases + deterministic extraction

MATCH
market demand / skill taxonomy

ANALYZE
gap + training impact

SIMULATE
baseline → scenario → delta

ARCHITECTURE

USER / FRONTEND
HTML • CSS • JavaScript

API LAYER
Python • FastAPI • Uvicorn

INTELLIGENCE ENGINES
Skill Gap • Training • Simulator

DATA LAYER
Jooble live + local fallback • taxonomy

PROTOTYPE SIGNALS

AUTO_FALLBACK
Live source → local data when unavailable

DETERMINISTIC
Repeatable skill extraction & scoring

API READY
REST endpoints for frontend integration

VALIDATED
Scenario baseline / delta tests

Current prototype ≠ trained employment predictor; verified outcome data is the next layer.

FEASIBILITY AND VIABILITY

WHY IT IS BUILDABLE

BUILDABLE NOW

FastAPI backend + working prototype engines

DATA-READY

Jooble + local fallback; taxonomy can be extended

SCALABLE NEXT

Add consented longitudinal outcome signals

RISK → CONTROL

Fragmented IDs

Stable consented record + identifier mapping

Incomplete outcomes

Confidence flags + multi-source validation

Privacy concerns

Consent, data minimization & revocation

API / data gaps

AUTO_FALLBACK + cached local prototype data

DEPLOYMENT PATH

Prototype → Verified datasets → Outcome tracking → Policy analytics

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IMPACT AND BENEFITS

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FROM TRAINING → OUTCOMES

ENROL
Training record

LEARN
Skills gained

PLACE
Employment signal

RETAIN
Retention / wage

IMPROVE
Course & policy action

REAL-LIFE SCENARIO 1,000 trainees finish a GIS course → SKILL PULSE can follow placement, retention and skill gaps by cohort and location.

TRAINEES
See missing skills & relevant pathways

TRAINERS
See which skills need reinforcement

EMPLOYERS
See regional talent availability

POLICYMAKERS
Compare programmes & interventions

Decision value: better outcome evidence • targeted training • stronger accountability • data-driven resource allocation

RESEARCH AND REFERENCES

RESEARCH BASE

SIH 2026 • PS SIH26135 — SIH 2026

Problem statement: employment outcomes, skill gaps & skilling impact

O*NET OnLine — onetonline.org

Occupation / skills taxonomy reference for skill–occupation mapping

Jooble REST API — Jooble API docs

Job-market signal source used by the prototype integration

Skill India Digital Hub — MSDE / SIDH

National skilling ecosystem reference / outcome-linking context

NOTE Current prototype demonstrates the workflow and scenario engine. Real-world employment outcomes require verified, consent-based longitudinal data.